Unit: Unit 9: Data Statistical Analysis
Topic: Stem and Leaf Plots

Name: ________________
Date: ____________

Quiz (Carolina Reaper)

Q1. A shipping company recorded delivery times (in minutes) for a week: $23, 19, 31, 27, 22$. What is the stem in a stem-and-leaf plot for these times?

(A) 1
(B) 2
(C) 3
(D) 20
(E) 30

Q2. In a stem-and-leaf plot for traffic flow data, the stem is 40 and the leaves are $2, 5, 8$. What are the actual data values?

(A) $40, 42, 45$
(B) $40, 402, 405$
(C) $40, 402, 408$
(D) $42, 45, 48$
(E) $402, 405, 408$

Q3. What does the leaf 7 represent in the stem-and-leaf plot with stem 50?

(A) 50.7 minutes
(B) 507 minutes
(C) 57 minutes
(D) 7 minutes
(E) 0.7 minutes

Q4. The stem-and-leaf plot shows travel times (in hours) for a set of trips: $3|2, 5, 9$. What is the median travel time?

(A) 3.2
(B) 3.5
(C) 32
(D) 35
(E) 3.9

Q5. Which data set has a stem-and-leaf plot with the most leaves?

(A) $10, 20, 30$
(B) $11, 12, 13$
(C) $101, 102, 103$
(D) $100, 200, 300$
(E) $1000, 2000, 3000$

Q6. The following data represents shipping times: $10.2, 10.5, 10.8$. Which is the correct stem-and-leaf plot?

(A) $1|0.2, 0.5, 0.8$
(B) $10|2, 5, 8$
(C) $10.|2, 5, 8$
(D) $100|2, 5, 8$
(E) $1000|2, 5, 8$

Q7. Which data set would be best represented by a stem-and-leaf plot?

(A) A small set of distinct values
(B) A large set of measurements with many repeated values
(C) A set of exam scores with a clear pass/fail threshold
(D) A set of categorical data with no natural ordering
(E) A set of numerical data with many decimal places

Q8. A delivery company recorded package weights (in pounds): $23.1, 19.5, 27.8$. How would the stem-and-leaf plot be constructed?

(A) Split at the decimal point
(B) Split after the first digit
(C) Split after the first two digits
(D) Do not split the values
(E) Only consider whole numbers

Open Ended Questions (FRQ)

Q9. Create a stem-and-leaf plot for the following set of data: $14.2, 15.1, 14.9, 16.3, 15.7$. Use 14 and 15 as stems. Calculate the median of the dataset.

Q10. Given a stem-and-leaf plot: $40|2, 5, 8$ and $50|1, 3, 6$, calculate the mean and range of the data represented. Show all work and explain your reasoning.

Chef's Tips

- Tip 1: Ensure students understand the concept of stem-and-leaf plots before attempting complex problems.
- Tip 2: Encourage students to use real-world examples, such as transportation data, to practice constructing stem-and-leaf plots.
- Tip 3: Remind students to consider the distribution of data and potential outliers when interpreting stem-and-leaf plots.